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Nobody owns the haze

An open warning-system prototype for Thailand: an audit of its first 110 days

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ABSTRACT

AirDash integrates public environmental feeds into a warning-system prototype for Thailand and neighbouring countries. This developer-led audit evaluates data coverage, collection reliability, retrospective forecast comparisons and corroborating optical signals; it does not test institutional coordination, public uptake or health effects. Over 110 days (17 July–3 November 2026), 58,298 scheduled pulls were logged, 98.7% as successful, although two token-gated feeds produced no readings. Feed timestamps reached the collector a median of 28 minutes after their labelled hour; notification delivery could not be verified from the available broadcast logs. In the observed ground-feed sample, 45% of stations were in Bangkok and three provinces had none. Two official products assigned different bands on 23% of paired province-days, mainly between the two cleanest bands; their spatial support and averaging semantics differ. 86% of fire cell-days in the specified mainland box were outside Thailand, which does not quantify their contribution to Thai pollution. Only 3 of 13 warned province-days exceeded 37.5 $\mu g\,m^{-3}$ at any station's derived daily mean. Persistence outperformed a retrospective CAMS correction variant; issue-time data availability was not reconstructed. Cameras and airport visibility offered weak corroboration in this predominantly wet-season record. We release summary inputs, corrected figures and typesetting code, identify limits to independent reproduction, and propose prospective tests requiring an implementation freeze before evaluation.

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1 Introduction

Every dry season the north of Thailand disappears under smoke. The cost is counted in clinics. In the upper north, same-day PM_{10} was associated with more outpatient visits for chronic lower respiratory disease (incidence rate ratio 1.020, 95% CI 1.012–1.028) [1]; all-cause mortality was associated with a 0.9% increase for every 10 $\mu g\,m^{-3}$ of vegetation-fire $PM_{2.5}$ [2]; and in haze periods the fatality and re-admission rates for COPD rose significantly [3]. Globally, smoke from landscape fires is estimated to kill about 339,000 people a year, 110,000 of them in Southeast Asia [4].

The smoke is also not only Thai. A source-apportionment simulation of 2019 for the upper north puts Myanmar as the main contributor in the west and centre and domestic burning in the east at the peak of the season [5]. Burma alone accounted for 37% of the hotspots in Southeast Asia in March in one analysis [6]. Across Southeast Asia, an estimated 23% of the roughly 900 thousand premature deaths a year from air pollution come from transboundary air, with impacts especially adverse in border regions [7]. Origin and destination are different places, run by different governments.

Southeast Asia has a treaty for this. The ASEAN Agreement on Transboundary Haze Pollution was signed in 2002 [8]; Indonesia, the last of the ten members, deposited its instrument in January 2015 [9, 10]. The agreement is a legally binding treaty; scholars criticise its limited enforcement and emphasis on sovereignty and non-intervention [11, 12], and two ASEAN roadmaps have followed, for 2016–2020 and for 2023–2030 [13, 14]. Inside Thailand, policy reviews describe many agencies with different mandates and little integration across sectors [15], and a structural problem of silo culture [16]; one analysis argues that fragmentation and protected economic interests have compromised the transboundary governance of the air [17]. We do not try to settle the diplomacy here. We take it as given that neither the treaty nor the agencies will change before the next burning season, and ask what can be built in the meantime.

We name three gaps. The first is *coordination*. The data that matter are public, but they sit in separate systems: ground stations at one agency, a satellite–ground fusion at another, burn scars at a research institute, weather at a third, and fires at NASA. The second is *responsiveness*. A number on a dashboard is not a response. Alerts can change behaviour but they also fatigue, the evidence on health outcomes is mixed, and localised, actionable information is likely to help [18–20]. The third is the *end-to-end chain*. Nobody owns the path from the field where the fire starts, through the air that carries it, to the person it reaches.

Research question. Can a small team integrate published environmental feeds into a usable warning-system prototype, and what limitations emerge from an audit of its initial operational record? Agency coordination and behaviour change are motivations, not outcomes measured here.

Contributions. (i) An open design and implementation of the chain (Section 3). (ii) A quantitative audit of the integrated record: 110 days, 175 ground stations and 58,298 pulls (Section 5). (iii) The system's failures, reported with the same weight as its successes. (iv) A proposed prospective evaluation plan, requiring executable protocols and author approval before the evaluation window (Section 6).

Scope. We have not obtained a multi-year, machine-readable ground record. The ground record here begins on 17 July 2026 and ends on 3 November 2026, before our proposed evaluation window of 1 December to 30 April; actual burning seasons vary geographically and meteorologically. The multi-year argument rests on the literature and on three seasons of burn-scar maps (2024–2026). What follows is therefore a systems paper with an early-season audit, not a trend analysis. Where the data cannot carry a claim, we say so.

2 Background

Where the smoke comes from. A 2003–2014 modelling study attributed nearly all visibility events below 7 km in its Southeast Asian sample to fire aerosols, and 99% of simulated fire-produced $PM_{2.5}$ in Bangkok [21]. The region's aerosol system is among the most complex in the world and its satellite aerosol products diverge widely [22]; field campaigns document the biomass-burning aerosol near its sources and far downwind [23]. El Niño strongly modulates burning in the Maritime Continent [24, 25], but Cambodia, Myanmar and Thailand respond far less: hotspot severity rose by under 25% in El Niño, positive-dipole or combined years, against more than 72% in

Kalimantan in the same comparison [26]. One global chemistry model underestimated the aerosol burden, most strongly early in the burning season [27].

What can be measured. Regulatory stations remain the reference. Low-cost optical sensors fill the gaps but read high without calibration, by about 40% in one large US evaluation [28], and their fitness for purpose depends on the calibration and on the aerosol [29–31]. More than 1,500 such sensors are installed in Thailand [32]; DustBoy, run from Chiang Mai University, is one such network [33]. From space, VIIRS detects active fires at 375 m [34] through NASA's FIRMS service [35, 36], and AERONET gives the column aerosol optical depth and the Ångström exponent, a qualitative indicator of particle size [37–40]. Thai agencies publish a fire layer [41] and a burn-scar monitoring site [42], and academic work maps burned area from Sentinel-2 imagery [43].

What can be forecast. CAMS, the Copernicus atmosphere service, is the free workhorse; its reanalysis assimilates satellite aerosol retrievals [44]. Its $PM_{2.5}$ forecast skill is modest and uneven: weaker over Malaysia than elsewhere [45], strongly lead-dependent in China [46], and biased in different directions in different regions [47]. Model output statistics, regressing the model onto local observations, is the classical remedy [48]. Trajectory models such as HYSPLIT trace where an air parcel came from [49].

What the eye and the camera can see. Horizontal visibility follows the Koschmieder relation between extinction and range [50], and visibility-derived extinction tracks $PM_{2.5}$ well in polluted regions but poorly in the cleanest [51]. Haze can be estimated from a single image [52], and camera-derived extinction has correlated with $PM_{2.5}$ at $R^2 = 0.7$ over a limited concentration range [53]. A hybrid CNN–LSTM on Bangkok photographs improved on plain CNNs [54], and a model that joins surveillance images to numerical data forecast $PM_{2.5}$ in Shanghai well [55]. In Chiang Mai, however, trained staff estimating the visual range reached 79% accuracy for PM_{10} but only 26% for $PM_{2.5}$ [56].

Warning people. Warnings change what people do, and they fatigue. A Sydney study found alerts cut cycling by 14–35% and also found evidence of alert fatigue [18]; English data show avoidance behaviour for children's asthma [57]; a Cochrane review found the evidence on health-care use conflicting [19]; and a review of 37 communication studies found actionable, localised information likely to help [20]. A review of citizen-science monitoring found little research on how its data feed policy [58].

3 The system

3.1 Design rules

We wrote five rules and have tried to keep them; Table 2 and Section 5 show where we did not. **Published interfaces first.** Integrate published feeds subject to their access conditions, attribution, tokens and quotas; public availability does not imply unrestricted reuse. These integrations did not require bespoke inter-agency data-sharing agreements. Where a stream needs a free token, the system is meant to say so rather than fail quietly. **Provenance on every number.** Each value carries its source, its observation time and its age; a number without a time is not shown. **Corroborate, never substitute.** Cameras, airport visibility and satellite aerosol may raise a flag or add a reason to the verdict; they never produce a $\mu g\,m^{-3}$ figure. **Fail closed, and say so.** Inbound webhooks are refused until their secrets are configured. "Unknown" is never reported as "fine". **One laptop.** The server is Node.js with 0 package dependencies and one SQLite file; anyone can run it. The code is 30,009 lines of server, 29,372 of browser JavaScript and 10,064 of tests in 97 files, with 85 API routes, over 349 commits since 2026-07-16, written with AI coding assistants under the direction of one author.

3.2 The chain

Figure 1 shows the path from a fire to a phone. Table 1 lists what the system reads.

Ground and fusion. The PCD-hosted feed supplied timestamped snapshots from 175 $PM_{2.5}$ stations; the Geo-Informatics and Space Technology Development Agency (GISTDA) publishes a satellite–ground fusion for all 77 provinces. Both are polled and stored with their own observation times.

Forecast and its correction. CAMS $PM_{2.5}$ forecasts arrive through Open-Meteo for each province. They read low against Thai ground stations, so the system applies a province-specific ratio learned over the trailing 14 days, clamped to 0.5–3, and falls back to the national ratio when a province has fewer than five paired days. This is a simple form of model output statistics [48]. A ratio learned on clean days overshoots on dirty ones, so it applies only up to the mean CAMS level it was learned on; beyond that, the calibrated absolute offset is added instead.

Where the smoke is coming from. VIIRS detections from three satellites (Suomi-NPP, NOAA-20, NOAA-21) are gridded to 0.25° cells. For each province the system sums the fire radiative power of cells within 500 km and within ±45° of the forecast wind-from direction, weighted by $1/(1 + d/150)$ for distance d in km, and grades the resulting proxy at 20, 150 and 600 index units. Missing or zero radiative power receives a unit fallback in the replay, so this index is not a calibrated MW total. This is a transport indicator, not a concentration forecast: it says where smoke can come from, not how much will arrive.

The verdict. A province's Air Watch Score is 0.40 ground $PM_{2.5}$ + 0.10 other pollutants + 0.15 trend + 0.20 forecast + 0.15 stagnation, anchored to the Thai air quality index breakpoints, whose 24-hour $PM_{2.5}$ values are 15, 25, 37.5 and 75 $\mu g\,m^{-3}$ [59]. Thailand's 24-hour standard has been 37.5 $\mu g\,m^{-3}$ since 1 June 2023 [60], which equals the WHO interim target 3; the WHO guideline is 15 [61].

Corroboration. Public cameras (about 1,000 in the catalogue, drawn from GISTDA, iTIC, the Department of Water Resources and a volunteer aggregator) are judged only against their own clear-day baseline at a similar hour. A province whose cameras look hazier than usual gets a reason added to its verdict; if fog is likelier (relative humidity ≥ 90% or an airport report of mist) the reason says so. Airport visibility from METAR reports is read through the Koschmieder relation [50]. In exactly one case do cameras move a band: no ground sensor is reporting and haze is widespread without fog, which raises the province to at least WATCH.

The push. Alerts are raised by rule (level crossed, forecast above the line, all-clear, rain washout) and fanned out to Telegram and LINE subscribers. Telegram is the main channel because the free LINE tier is capped at 300 messages a month, which one haze episode would exhaust. A gap of 30 minutes between broadcasts and of three hours per subscriber keeps a volatile hour from becoming a flood. Both webhooks verify who is speaking.

3.3 Built to survive a season

The system runs on one laptop behind a named tunnel and a static front end. Table 2 lists what broke between July and October and what was changed.

Each row is a rule the system now enforces rather than a lesson learned.

4 Data and methods

Window. All analyses are frozen at 00:00 on 4 November 2026 (Bangkok time). Ground and reliability records cover 17 July to 3 November (110 days, 2,640 hours). Fire detections cover 21 October to 3 November (14 days); camera frames 28 September to 3 November; airport reports 11 October to 3 November; the alert and Telegram log 28 September to 3 November; the PCD reading lag from 7 August; burn scars for 16 months between January 2024 and April 2026 (62 provinces); forecast pairs 7 August to 3 November.

Ground record. Values are snapshots of the PCD-hosted feed. Their underlying averaging period was not retained, so hourly timestamps do not establish independent one-hour exposures. A station-day needs at least 18 snapshots; a province-day averages those station-day means. These derived means are compared with guideline thresholds, not certified as official regulatory exceedances. Values at or above 500 $\mu g\,m^{-3}$ are excluded by a screening rule (11 values at 2 stations; raw maximum 950). Extreme concentrations can be real; this threshold does not diagnose instrument faults, and sensitivity to the exclusion has not been established. An exceedance is labelled isolated when at least two stations within 50 km reported that day and their median was under half its value; this cannot distinguish a local source from measurement error. "Upper north" means Chiang Mai, Chiang Rai, Lampang, Lamphun, Mae Hong Son, Nan, Phayao and Phrae.

Intervals. Where the unit of analysis repeats (frames per camera and hours per airport; fire comparisons by calendar day) we use a 95% percentile interval from a bootstrap that resamples whole clusters [62], with 400 to 2,000 resamples depending on cost. Simple proportions over station-days, detections, alerts and warned province-days carry Wilson 95% intervals [63, 64] that treat the observations as independent; these are optimistic where events cluster in time, and we say so where it matters. Rank correlations are Spearman's ρ [65]; agreement between category assignments is Cohen's κ.

Forecast verification. For lead k, the target is a derived province-day mean on day D and the forecast is the first selected morning issue on local day $D - k$. This is a retrospective calendar-day holdout variant, not a reconstructed operational replay. Correction ratios use completed daily means through $I-1$, and persistence uses the previous day's mean; their availability at issue time was not checked using receipt timestamps. The replay differs from production's issue aggregation and treatment of partial current-day observations. Forecast intervals resample provinces, not calendar days, and do not account for shared daily shocks or serial dependence. They are exploratory; an as-of replay and day/block sensitivity are required before operational skill claims. Events use POD, false-alarm ratio and CSI [66, 67].

Smoke transport replay. The upwind indicator is re-implemented from the production code (500 km radius, ±45° sector, distance weights, 0.25° cells, a two-day look-back, cells within 15 km counted whatever the wind) and restricted to the box the production code uses, 4–29°N and 92–110°E. It is evaluated with the wind forecast issued on the morning of each day (today's, where production prefers tomorrow's). Each province's next-day $PM_{2.5}$ and log weighted source index are centred on the province's own mean, to remove geography, and then on the day's mean across provinces, to remove everything that changes for all provinces together, such as the fortnight-long fall in fire counts. Provinces share the weather on a given day, so the interval resamples days, of which only 9 have a next-day record.

Reproducibility. Data-derived template statistics are filled from frozen JSON summaries; missing or non-finite values stop the build. The supplement includes these summaries, selected extracts, figure PDFs, manuscript source and build code, sufficient to rebuild the typeset PDF with Python and LaTeX. It does not release the complete databases or operational logs: several extraction scripts still require them, so independent end-to-end statistical reproduction is not yet possible. Operational analysis uses aggregate alert/broadcast timestamps and counts, not subscriber identifiers, subscriber-authored message contents or behaviour.

5 Results

5.1 The network we inherited The observed PCD-hosted feed contained $PM_{2.5}$ snapshots from 175 stations in 74 of 77 provinces (Fig. 2a). Bangkok holds 79 of them, 45% of the network. Three provinces (Phetchaburi, Phang Nga and Ranong) have none in this feed sample, and 60 have exactly one. The eight upper-north provinces, where the dust season does its worst, share 14 stations. Only 67 stations also report PM_{10}, in 36 provinces, so size-fraction comparisons are available in fewer than half the provinces; these alone cannot identify smoke, dust or chemical composition. Of the 161 stations that produced at least one full station-day, 132 reported at least 90% of their hours, and 17 reported fewer than half.

The record itself is quiet (Fig. 3). Across 14,357 station-days, 38% exceeded the numerical WHO 24-hour benchmark of 15 $\mu g\,m^{-3}$, but only 31 (0.22%; 95% CI 0.15%–0.31%) exceeded the numerical Thai 24-hour benchmark of 37.5, and 2 exceeded 75. The monthly mean of station-days was 14.2 in July (from the 17th), 11.4 in August, 13.0 in September and 17.3 $\mu g\,m^{-3}$ in October, with a station mix that changes from month to month. The worst day was 9 October, when the highest station in each of 6 provinces crossed 37.5. Of the 31 exceeding station-days, 12 were isolated spikes, 13 were not isolated, and 6 had too few neighbours to tell. The largest value in the record, 151.8 $\mu g\,m^{-3}$ at one station in Chon Buri on 7 September, stood alone: the median of its 7 neighbours within 50 km was 12.9. That station supplied 8 of the 11 screened-out values, and the highest hourly value that survived the screening filter was 461. We read this as a baseline. It is the clean air against which the burning season will be judged, and it is not evidence that the north is safe.

5.2 Does the machinery hold? The collector made 58,298 scheduled pulls across 23 sources, and 747 failed (1.3%). It was up 94.6% of the window. Seven gaps longer than six hours add up to 122 hours (Fig. 4a). Two match entries in Table 2: 19.8 hours on 26–27 September (the database incident) and 30.9 hours on 24–25 October (the disk failure). The other five, the longest 34.3 hours on 23–25 August, have no entry in our operating record and we cannot explain them. The newest successful PCD pull was no older than two hours 95.3% of the time; the feed was stale for 124 hours in 19 episodes.

The timestamp comparisons describe separate processes (Fig. 4b–d). A PCD snapshot was first seen a median of 28 minutes after its labelled hour (95th percentile 67). For 485 severe alerts, the next logged broadcast occurred a median of 7 seconds later. This is only a next-broadcast timing proxy: alert

Table 1. What the system reads. Seventeen streams, their owners, interface and measured behaviour over the 110-day window. "Interval" is the configured polling interval, not a measured one. "Pulls" counts scheduled attempts; "success" is the share the adapter logged as successful, which is not the same as delivering data (see text).

Stream	Owner	Interface	Interval (set)	Pulls	Success
PM2.5, PM10, gases, 175 ground stations	Pollution Control Dept. (PCD)	JSON	15 min	3,865	99.3%
PM2.5 fused with satellite, 77 provinces	GISTDA	JSON	1 h	2,549	98.1%
Noise level, 27 stations	PCD (Noise4Thai)	HTML page, parsed	30 min	5,038	99.4%
Rain gauges, about 4,200	HII, for several agencies	JSON	10 min	14,701	99.1%
Weather forecast and stations	Thai Meteorological Dept.	JSON via sibling relay	15 min	4,235	98.2%
Wind, rain and PM2.5 model forecast	Open-Meteo (CAMS, weather models)	JSON	6 h	1,657	92.2%
Satellite rain (IMERG)	NASA GES DISC	OPeNDAP	30 min	5,077	100.0%
El Niño index (ONI)	NOAA CPC	text file	12 h	342	94.2%
Agricultural burn scars by crop	HII and Kasetsart Univ.	JSON	daily	105	97.1%
Drought risk, crop water use	GISTDA	JSON (GeoJSON)	6 h	230	98.3%
Fire detections, three VIIRS satellites	NASA FIRMS	CSV	6 h	196	92.9%
Aerosol optical depth, 5 sites	NASA AERONET	CSV	daily	46	97.8%
Airport visibility and weather codes	NOAA Aviation Weather Center	JSON	30 min	1,578	97.9%
Low-cost PM2.5 sensors, northern Thailand	Chiang Mai Univ. (DustBoy)	JSON	1 h	831	100.0%
About 1,000 public cameras	GISTDA, iTIC, DWR, volunteer aggregator	HLS, MJPEG, JPEG	30 min	3,370	99.1%
Haze news and citizen reports	News outlets (RSS)	RSS	20-30 min	7,474	99.1%
Flood verdicts (sibling system)	FloodDash	JSON, local	10 min	6,398	98.5%

Table 2. What broke, and the rule that now prevents it. Dates are 2026. Entries are taken from the project's operating record; the last was found while preparing this paper.

When	What happened	What is enforced now
19 Sep	Another local service bound the API's port and the tunnel served it; the public API was dead for about 20 hours.	The port moved; every response carries a service header; the edge and the watchdog treat a response without it as "backend down".
26–27 Sep	The live database was placed on a USB hard disk; synchronous queries froze the site for 4.5 hours (the collector log shows a 19.8-hour gap on 26–27 September).	The live database stays on the SSD as a bounded hot tier; the permanent record is archived to the external drive.
27 Sep	One syntax error in a browser module left every visitor on the boot splash while server health stayed green.	Deployment parses every browser module first and refuses to ship on error.
27 Sep	Thirteen restarts in one day spent the shared Open-Meteo quota.	A source that succeeded within its interval waits for its normal turn after a restart.
28 Sep	A bias ratio of 2.9 (Rayong) turned a raw forecast of 14 into "41", crossed the warning line and was broadcast.	The ratio applies only up to the CAMS level it was learned on; beyond that, an absolute offset.
24 Oct	The internal disk filled; log writes failed and the machine died, 30.9 hours of downtime.	A guard measures free space and sheds optional writes (camera frames) below 3 GB; readings keep running.
Nov	Found while preparing this paper, after the airport data analysed here were collected: 10 of 28 airport codes in the visibility table were dead and 3 were mislabelled.	The table was rebuilt from the aviation weather API; a verifier script checks it against the live service.

membership in that batch and successful recipient acceptance were not linked. It cannot establish alert hand-off, delivery, or end-to-end latency. We withdraw the simulated combined-hop latency from the earlier version because independent random pairing does not reconstruct a real alert chain. VIIRS detections were first seen a median of 392 minutes after overpass, with a tenth exceeding 23.6 hours. Six-hour polling contributes to this lag, alongside upstream publication and availability; their contributions were not separated. The interface should therefore display observation age rather than imply a live fire picture.

Success is not delivery. Five sources wrote no new readings in 110 days. Three of them (the weather relay, camera health and the flood sister system) relay into a key–value store or other tables by design. The other two, satellite rain (IMERG; 5,077 pulls) and DustBoy (831 pulls), are gated by a free token that nobody had registered; their adapters skip quietly when it is absent, and both logged 100% success. Together they are 10% of all pulls, so the headline success rate flatters the system. The bureaucracy we set out to bypass came back at the smallest scale, as a registration form, and the logs reported it as success.

5.3 Two official answers to one question GISTDA's spatial fusion and sampled ground stations provide related but non-equivalent estimates; differing footprints, averaging periods and station mixes can produce disagreement. Over 5,748 paired province-days in 68 provinces they correlate (ρ = 0.76; 95% CI 0.72–0.79) but differ in level: GISTDA reads 2.0 µg m^{-3} higher on average, with 95% limits of agreement from −4.8 to 8.7 (Fig. 2b). Sorted into Thai air-quality bands, they place a province in the same band on 77% of days (κ = 0.49; weighted κ = 0.52), and GISTDA is the higher on 18% (Fig. 2c). Of the 1,314 disagreements, 88% lie between the two cleanest bands (15 or below, and 15–25), which GISTDA's offset largely explains. On the 5,748 province-days compared, PCD alone put a province above 25 on 51, GISTDA alone on 102, and both on 42. Above 37.5, GISTDA never did so; PCD did on 7 province-days. A citizen looking up the same province on two official sites will sometimes get two colours, usually adjacent ones. The two sources measure different things at different scales; we found no published rule for choosing between them.

5.4 The forecast CAMS read low against the ground. At the same-day lead its mean bias was −4.9 µg m^{-3} over 4,199 province-days (mean absolute error, MAE, 6.2), and the pooled ratio of observation to model was 1.62. The retrospective correction variant reduces descriptive error: MAE falls to 3.8, a change of −2.4 (95% CI −2.8–−2.0) (Fig. 5a–c). Within this retrospective sample it improves on raw CAMS but does not

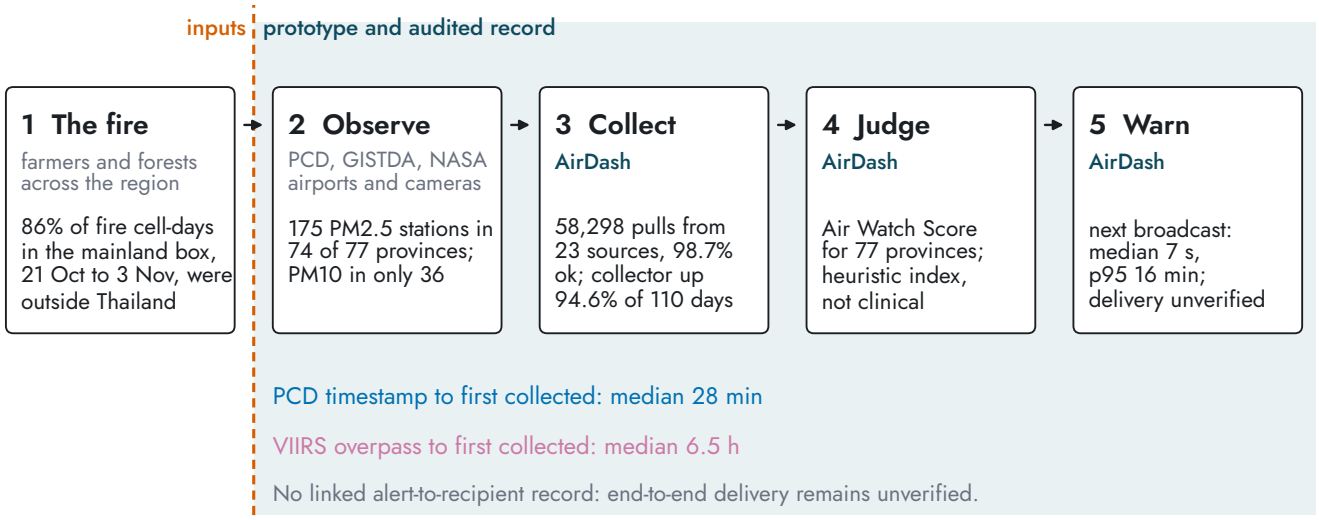


Fig. 1. From a burning field to a phone. The intended five stages. Collection delays are measured separately; the notification link is not verified end to end. Medians are from stored timestamps over the window; the satellite hop is the lag between a VIIRS overpass and the first time the system could see the detection.

beat persistence; issue-time availability and shared-day uncertainty remain unresolved. Carrying the last observed daily value forward has MAE 2.4 at the same-day lead, so the corrected model is 1.45 worse (CI 1.24–1.68). The gap narrows with lead (to 0.63 tomorrow and 0.50 the day after) but does not close. The correction leaves a residual bias of −1.2. The ratios behind it are large and uneven: the median province ratio was 1.9, and 14% of province-days sat at the clamp of 3.

For events the picture is thin because the season was quiet. At the same-day lead the corrected forecast detected 27 of 89 days above 25 (POD 0.30, FAR 0.72, CSI 0.17), against 0.29, 0.67 and 0.18 for persistence, and none of the 5 days above 37.5.

The system also logged forecast warnings; the alert table does not establish their dispatch. Of 21 forecast warnings in 7 provinces, 13 distinct province-days could be checked against ground data. The median warned value was 41 $\mu g\,m^{-3}$; the median observed province mean was 25. Only 3 of 13 reached 37.5 at any station (Wilson 95% CI 8%–50%) and 1 in the province mean (Fig. 5d). 10 of 13 over-called, though not wildly: 6 reached 25 in the province mean. All 21 warnings were logged between 30 September and 31 October (Bangkok time), after the bounded correction of Table 2, and the 13 province-days fall on only 6 target dates, so the interval above, which treats them as independent, is optimistic. We have not tested why they over-called.

5.5 Whose fire is it? In the 14 days from 21 October the detection feed held 196,385 detections, but 93% of them fell outside the box the production smoke indicator looks at, mostly in Indonesia, which the feed has covered since 19 October. In the production box (4–29°N, 92–110°E: mainland Southeast Asia and its neighbours) there were 13,779 detections, 1,348 of them inside Thailand. Three satellites often see the same fire, so we count 0.25° cells per day: 86% of the 5,159 fire cell-days lay outside Thailand (95% CI over days 84%–89%), and 91% of the fire radiative power (89%–93%) (Fig. 6b). The box also holds Laos, Cambodia, Vietnam, southern China, Peninsular Malaysia and Myanmar, so "outside" is not "Myanmar". On 73% of 847 province-days the weighted source index reached 20 units. The foreign-share calculation uses unweighted cell radiative-power proxies allocated by within-cell detection counts; it is not the foreign fraction of the distance-weighted index, a measured cross-border power total, or an attribution of Thai PM. Figure 6c retains this exploratory allocation with that limitation explicit.

Does the indicator predict anything? Across provinces and days, next-day $PM_{2.5}$ rose with the log of the weighted source index after removing each province's mean and each day's mean (ρ = 0.28; 95% CI over days 0.07–0.49; 527 province-days in 59 provinces, but only 9 days). In the eight upper-north provinces, where the claim matters most, it is ρ = 0.53 (CI 0.10–0.73); elsewhere ρ = 0.24 (CI 0.01–0.46). The result is not stable. Centring on the province alone gives ρ = 0.07 in the upper north, and eight provinces that share one wind are not eight independent tests. Nine days cannot settle it, and the window sits before the season in which it matters. We therefore do not claim that the indicator forecasts transported smoke; it describes where smoke could come from.

The domestic picture is steadier, for one kind of burning. The burn-scar product covers agricultural fires (rice, sugar cane and maize) in 62 provinces over three seasons; it does not map forest or grass fires. Agricultural burning is concentrated: ten provinces hold 52%, 60% and 53% of the burned area in 2024, 2025 and 2026, and the ranking barely moves between years (ρ = 0.96 for 2024 against 2026; CI 0.92–0.98); 8 of the ten are the same in all three years (Fig. 6a). The upper north holds only 6%–7% of the agricultural burned area of January to April, and there the crop is mostly maize (57%), where elsewhere it is paddy and sugar cane. How much of the upper north's smoke comes from forest fire instead, these maps cannot say. Agricultural burned area in January to April 2024, a season that followed an El Niño (the index stood at +1.84 in December–February, NOAA), was 23,454 km^2, 1.7 times that of 2025 (14,004 km^2); with three seasons, that is a hint and not a test.

5.6 What cameras and airports can see Public cameras are cheap and numerous, and of little use for this question so far. Of 114,803 frames from 993 cameras, 38% were taken in low light and 38% could be judged against a camera's own clear-day baseline. Across 21,535 frames from 104 cameras with a ground station nearby, contrast loss rose with $PM_{2.5}$ (Fig. 7a) but the correlation is weak: ρ = 0.11 (CI 0.07–0.15), and within a single camera over time the median ρ is 0.04. As an alarm, scored against the hourly $PM_{2.5}$ at the nearest station, the "possible haze" flag has specificity of 97% and sensitivity of 4% for frames at 25 or above, and precision of 11% (Fig. 7c). Of the 757 flagged

frames with a humidity reading, 39% were at 90% relative humidity or above, where fog is the likelier cause; a 1-in-25 sample of 1,205 unflagged frames gave 38%, so humidity does not separate them.

Airport visibility fares no better. METAR visibility is censored at 10 km in this dataset: 81% of hours provide only a lower bound on range, limiting resolution of cleaner conditions. Below the ceiling, visibility and $PM_{2.5}$ are weakly related, and with the opposite sign to the simple expectation that visibility decreases as particle extinction increases (ρ =+0.18; CI −0.07–0.24; 14 airports). In the 25–37.5 band, 64% of hours had visibility under 10 km against about 20% in the cleaner bands, but that rests on 270 hours at 5 airports (cluster CI 25%–72%), 76% of them from a single airport; without it the share is 33% (Fig. 7b). In 6,543 airport-hours the observers' weather codes never once attributed reduced visibility to haze or smoke, but those codes are normally reported only at low visibility and just 2.2% of hours were below 5 km. The test needs the burning season.

5.7 Alerts people would see The alert log holds 497 level alerts: 91 from PCD-hosted snapshots and 406 from GISTDA. Among 85 PCD and 399 GISTDA alerts with at least four available snapshots in the alert-hour-plus-five-hour window, 74% and 42%, respectively, had at least three snapshots at or above 37.5. This window includes the alert hour, not six wholly subsequent hours. The earlier daily-hit percentages are withdrawn: missing complete-day outcomes were counted as non-hits, and a same-calendar-day mean includes time before the alert. These outcomes need separate denominators and cannot establish prospective daily warning accuracy. Neighbour corroboration was 90% among 70 alerts with sufficient neighbours; 35% of PCD alerts came from five stations. Repeated alerts and shared stations are not independent evidence, and user fatigue was not measured.

5.8 The year it arrives in The Oceanic Niño Index for August to October 2026 is +2.46, the highest of the 77 August–October seasons in NOAA's record, above 1997 and 2015. Mainland Southeast Asia responds less to El Niño than the Maritime Continent does [26], and we make no forecast of the coming season from it. We note it because the proposed tests below will be read in a year that is unusual on one dimension.

6 Proposed prospective tests

This initial audit motivates the proposed tests in Table 3, for an evaluation window of 1 December 2026–30 April 2027. This is not an executable preregistration: current scripts contain fixed historical windows, and T1 lacks alert-to-batch identifiers. Before evaluation, authors must approve the protocol, specify missing-data denominators and averaging semantics, implement issue-time joins and the past-six-hour T2 rule, and freeze versioned code. Thresholds informed by this audit are exploratory; changes and failed tests must be published. We make no promise that the present scripts can be rerun unchanged.

7 Discussion

The audit identifies integration and verification problems within this prototype. It does not establish that agencies fail to coordinate or that the system improves public response.

Coordination. The integrated products have different spatial support and do not always agree. Two official sources assign a province to different air-quality bands on 23% of province-days; the ground network puts 45% of its stations in one city and 14 in the eight provinces the dust season hits hardest; seventeen streams (sixteen public feeds and one sister system) arrive through eight kinds of interface from more than ten organisations, two of them gated by a free account. None of this is a failure of any one agency. This audit does not establish the institutional causes of those differences. A small team could do the join, using published interfaces under their access conditions. That is the argument for the approach, and it is also its limit: a join can show the disagreement but cannot say which source is right.

Responsiveness. Speed is not the bottleneck; judgement is. Snapshots typically reach the collector in about half an hour; alert acceptance and delivery have not been verified. But the alert rule fires on hours, people respond to days, and 10 of 13 forecast warnings did not exceed the numerical benchmark at any station's derived daily mean. Alert fatigue is a documented cost [18], and an alert that is quick and often unconfirmed invites it. Table 3 proposes testing more selective alerts; the effect on people remains unknown.

The chain. The slowest measured link is the first, the fire detection (6.5 hours at the median), and much of that delay is our own six-hour polling, so the first fix is ours. The upwind indicator turned out to be what we said it was, a map of where smoke can come from, and not yet a forecast. The most useful thing in the chain may be the least glamorous: a dated, source-tagged statement that 86% of the fire cell-days on the mainland in a fortnight lay outside Thailand.

Bypassing, honestly. What did bypassing the bureaucracy buy? It bought speed (a working chain in one season, by one author with AI assistants and one laptop), implementation simplicity (no declared server npm dependencies; vendored libraries, hardware, hosting and labour still have costs, which were not evaluated) and a record of its own failures that nobody had to approve. It did not buy the sensors that are missing, the agreement between sources, or the registration form for a free token. Nor does it touch the diplomatic layer. Satellites can say that a fire burns in a particular place; they cannot say who lit it, and we make no claim of attribution. What an open record can do for diplomacy is modest: it offers a shared set of numbers on which an argument could start. The treaty rests on each party's own policies for its own territory [8], and whether such a record changes anything is not something this paper tests.

What we would ask of others. These are proposals that follow from the audit, not findings. (i) Agree one published value per province, or publish the difference between sources with the reason. (ii) Fill the upper-north gap with calibrated low-cost sensors [28, 32] whose record is published beside the regulatory one. (iii) Pair every $PM_{2.5}$ station with a PM_{10} reading. (iv) Warn on persistence, not on the hour. (v) Publish the age of every fire detection beside it. (vi) Say what happens to a free-token source that has no token. None needs a treaty.

8 Limitations

The ground record is 110 days long and ends before the dust season, so we cannot say how the system behaves in the conditions it was built for. Most conclusions about burning rest on a 14-day window of fire detections and on three seasons of burn scars; the one forecast we could verify, CAMS, is the only model we tested, and we did not run trajectory models [49]. The correction, the thresholds and the weights of the score were set by the authors and not fitted to health outcomes, which we did not study. Station-day means rest on the PCD feed as served, and while we screened out values of 500 µg m^{-3} or more, we did

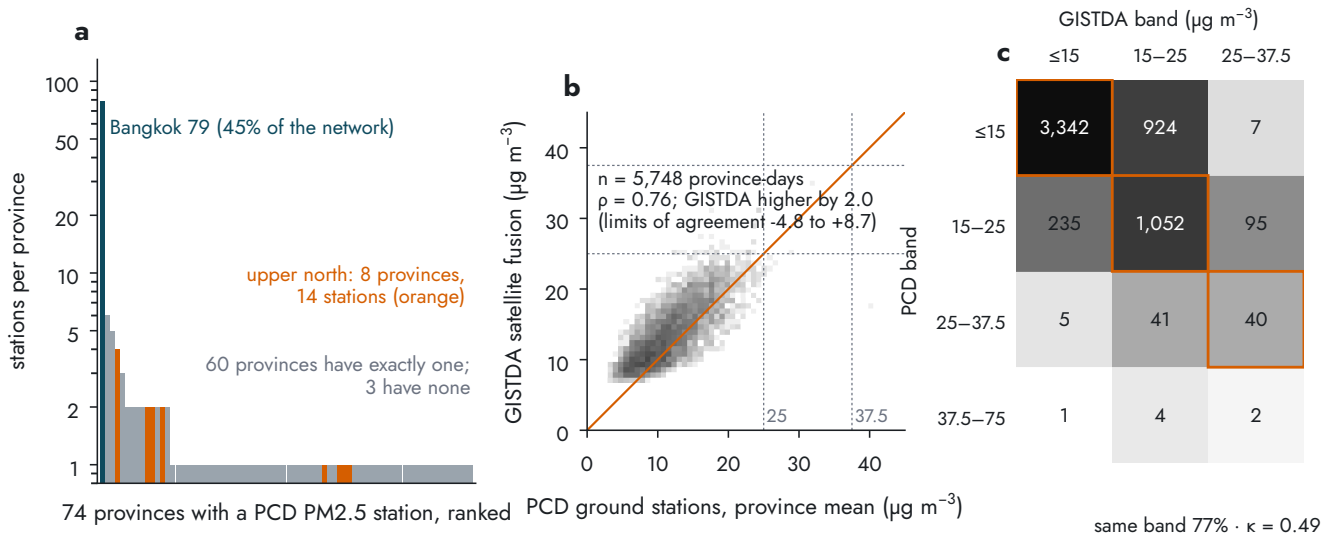


Fig. 2. Where the sensors are, and two official answers. a, Stations per province among the 74 provinces with a PCD $PM_{2.5}$ station (log scale); upper-north provinces in orange. **b**, Province-day means from PCD stations against the GISTDA satellite–ground fusion; the line is equality, dotted lines mark 25 and 37.5. **c**, The same province-days sorted into Thai air-quality bands; outlined cells are agreement.

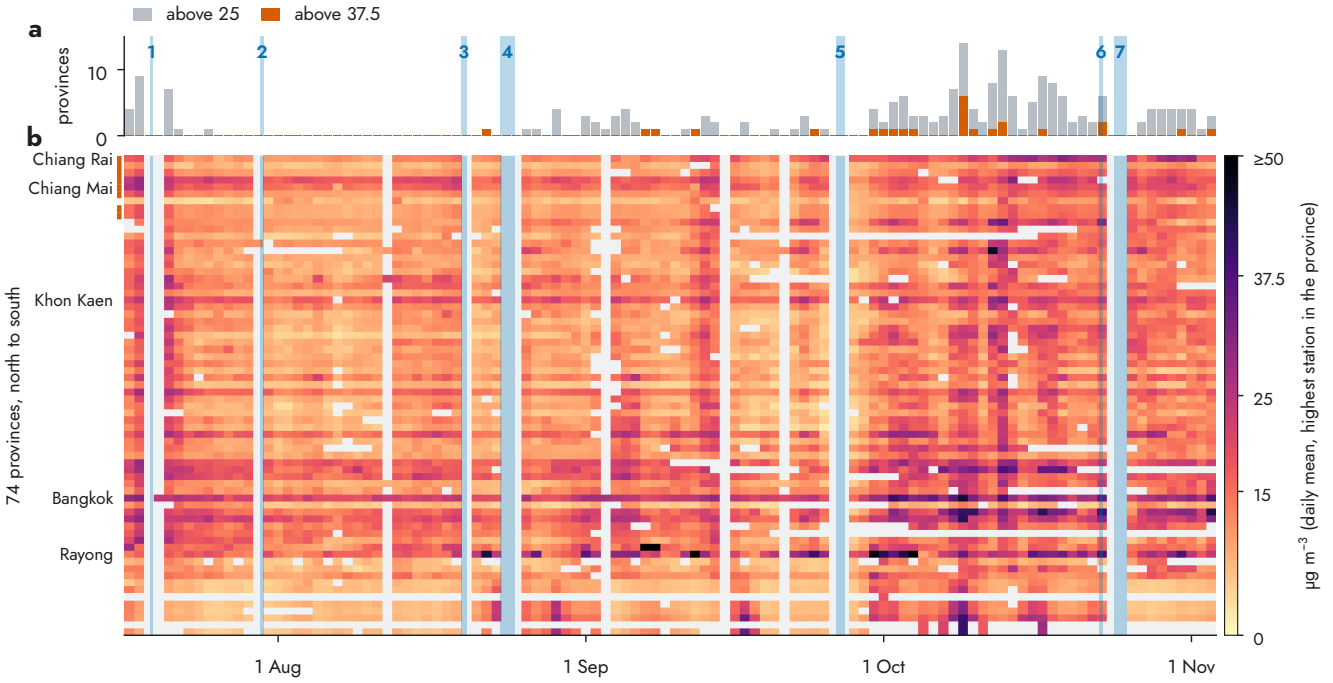


Fig. 3. The 110-day record. a, Provinces whose highest station-day mean exceeded 25 (grey) or 37.5 (orange) each day. **b**, Highest station-day mean per province and day, provinces ordered north to south. Pale blue bands are collector outages of over six hours (numbered as in Fig. 4a); grey cells are missing data. Orange ticks at left mark upper-north provinces.

not validate that exclusion or audit calibration. The verification sample for high events is small: 5 province-days above 37.5. The push channel was exercised with one or two subscribed chats, and the broadcast-timing proxy does not establish server acceptance or delivery to any person. Many of the intervals use Wilson bounds that treat events as independent when they cluster in time, and the fire and transport results rest on 14 days, of which nine have a next-day record. The airport data were collected before the visibility station table was rebuilt, when 3 of 28 codes were mislabelled. Camera and visibility analyses are pilots; their confidence intervals are wide and one airport dominates the visibility result. The system runs on one machine and its record contains its own gaps; five gaps of over six hours are unexplained. The code was largely written with AI assistants, which is a risk we reduced with tests (97 files at the frozen code census) and did not remove. We have not shown that the system changes anyone's behaviour or health, or that people can use it right away beyond a handful of test users. Finally, the audit was prepared by the system's developers; the tests above, and the public code, exist so that others can check it.

Declarations

Data and code. The system and analysis code are available at https://github.com/Nonarkara/airdash, under paper/; the derived data behind every figure and every number in the text are in paper/data, and the preprint is rebuilt from them with paper/build.py. The live database and archive that the extraction step reads are not released. The live system is at https://air.nonarkara.org. Upstream data are the public feeds of the organisations in Table 1.

Use of AI. The analysis code, the figures and the first draft of this text were produced with Claude (Anthropic) under the authors' direction,

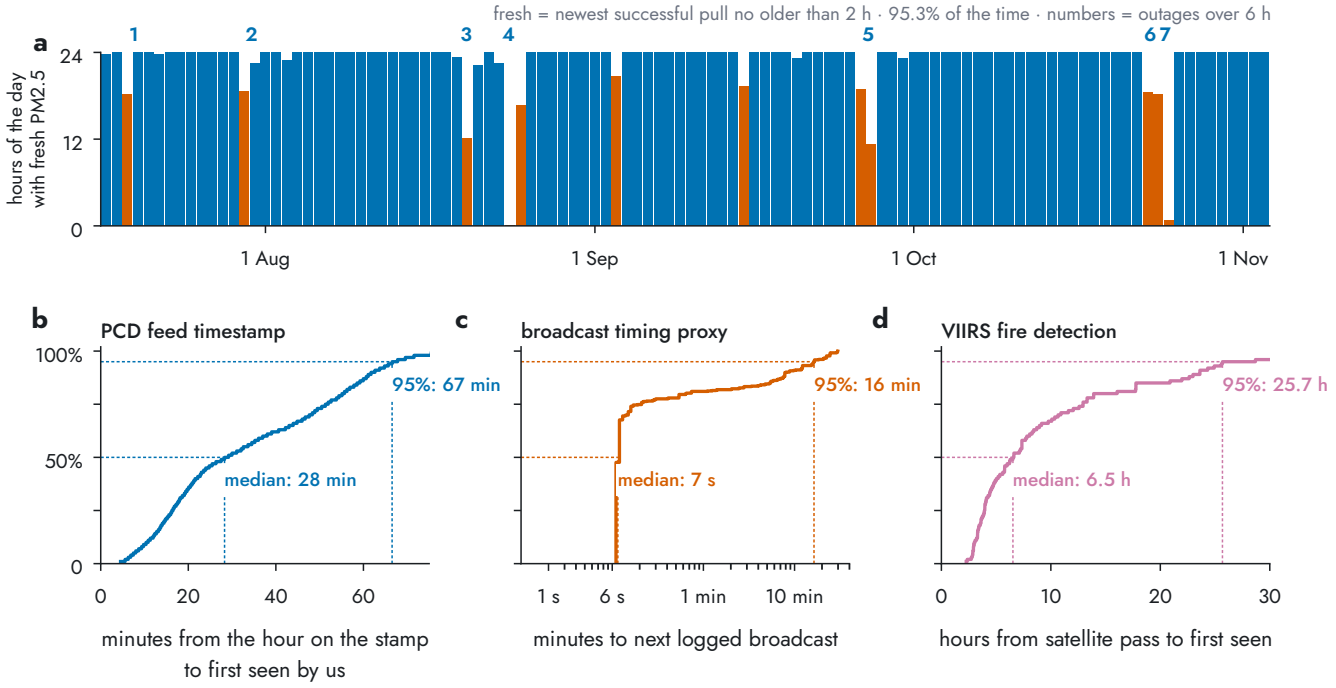


Fig. 4. Does it stay up, and how long is the wait? **a**, Hours of each day with a fresh PM2.5 feed (newest successful pull no older than two hours); orange days had a fresh feed for under 90% of the day and numbers mark the seven outages over six hours. **b–d**, Empirical distributions of the three clocks: the lag from the hour on a PCD reading to the first time the system saw it, the time from an alert to the next logged broadcast (batch membership and delivery unverified), and the lag from a VIIRS overpass to first sight of the detection.

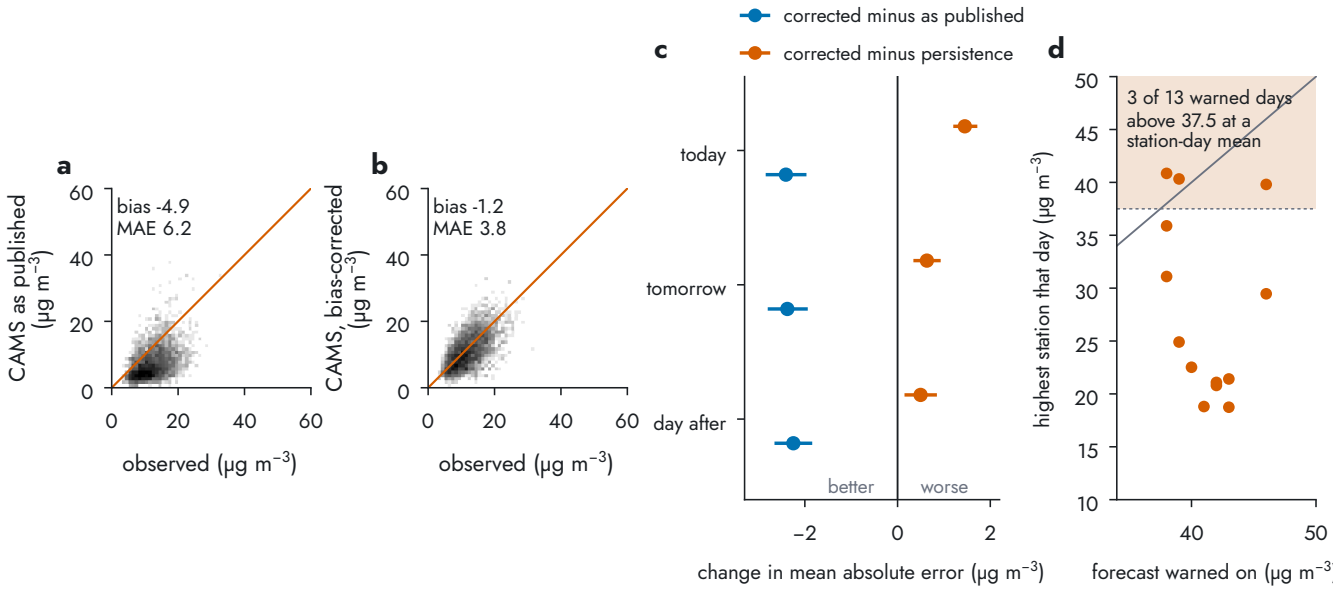


Fig. 5. The correction helps, persistence still wins, and the warnings over-called. **a, b**, Same-day CAMS forecast against province-day observations, as published (a) and after the retrospective correction variant (b); line is equality. **c**, Change in mean absolute error with exploratory 95% province-cluster intervals, by lead; left of zero is better. **d**, The 13 distinct province-days on which a forecast warning was logged: the highest value warned for the province-day against the highest station that day; the shaded region is where the derived station-day mean exceeded the numerical 37.5 benchmark; source averaging periods remain unverified.

and the system's code was written with AI coding assistants. Every number is computed from the stored record by the code in the repository. References were checked against Crossref or the source's own page; for some, only the bibliographic record could be verified. An independent automated review found errors in the first version, which were corrected. The authors are responsible for the content.

Competing interests. N.A. developed and operates AirDash, which is free to use and carries no advertising; depa, N.A.'s affiliation, is named as a steward of the project.

Author contributions and funding. To be completed by the authors.

Ethics. Environmental observations and aggregate operational alert/broadcast timestamps and counts were analysed. No subscriber identifiers, subscriber-authored message contents or behaviour were analysed; no ethics approval is claimed.

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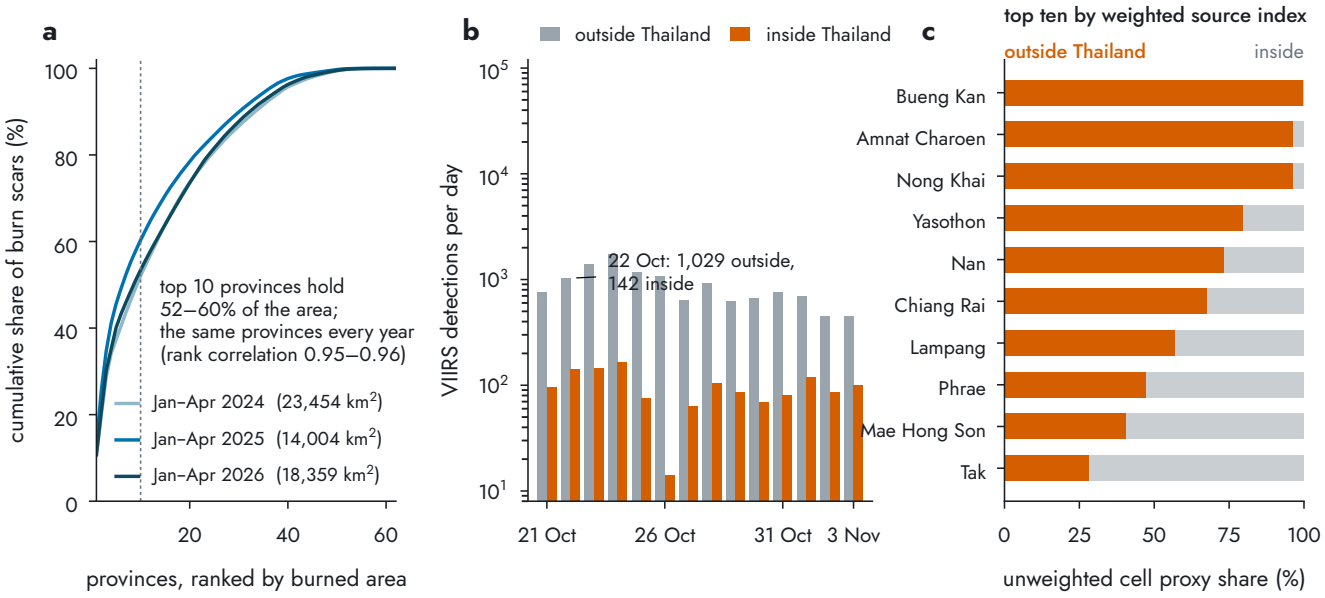


Fig. 6. Where the fire is. **a**, Cumulative share of burn-scar area by province, ranked, for January–April of three seasons. **b**, VIIRS detections per day (log scale) inside and outside Thailand in the production box (4–29°N, 92–110°E), 21 October to 3 November. **c**, For the ten provinces with the highest weighted source index, the outside-Thailand share of an unweighted cell-allocation proxy (not the weighted transport index).

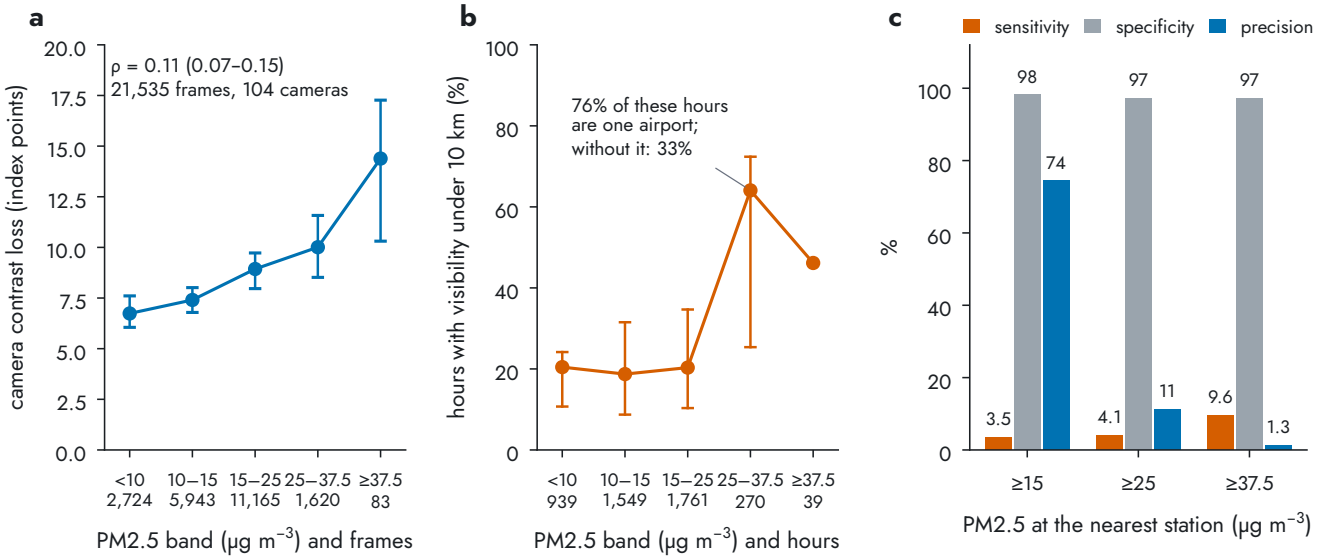


Fig. 7. What cameras and airport visibility can and cannot say. **a**, Camera contrast loss (a relative index, not a concentration) by PM2.5 band at the nearest station; bars are 95% cluster-bootstrap intervals over cameras. **b**, Share of airport-hours with visibility below 10 km, by band of PM2.5 at the nearest station. **c**, Sensitivity, specificity and precision of the "possible haze" camera flag at three thresholds.

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Table 3. Proposed tests for the 2026–27 evaluation window; implementation and author approval pending. Baselines are this paper's results. Proposed "pass" criteria are stated here; "if not" is what the system does about a failure. CI is a 95% interval; CAMS is the bias-corrected forecast.

	Question	Baseline (this paper)	Pass	If not
T1	Is each eligible severe alert accepted by Telegram within 35 minutes?	Not established; next-broadcast timing does not verify membership or acceptance	At least 95%, linking alert IDs to batch IDs and successful recipient responses; include failed/unmatched alerts	Diagnose queue and send failures before changing batching
T2	Do persistent alerts predict a subsequent daily exposure window?	Not established; prior same-day outcome rates had invalid missing-data denominators	Define a forward 24-hour target and completeness rule; compare all alerts with a past-six-hour persistence rule; proposed lower CI limit 50%	Retain a watch tier; do not present hourly snapshots as a daily exposure forecast
T3	Does the corrected forecast beat carrying the last observation forward two days ahead?	MAE difference +0.50 (CI +0.19–+0.81), worse than carrying forward	Upper CI limit of the difference below zero	Stop issuing the second-day warning; show it as a range, not a value
T4	Does upwind fire predict next-day $PM_{2.5}$ in the upper north, February–April?	$\rho = 0.53$ (CI over days 0.10–0.73; 9 days), 0.07 without day-centring	$\rho \geq 0.25$ with the lower CI limit above zero, over at least 60 days	Describe the indicator as a map of sources only, and drop it from the verdict
T5	Do cameras track $PM_{2.5}$ within a camera over time?	Median within-camera $\rho = 0.04$	Median $\rho \geq 0.30$ among cameras with at least 20 judged frames	Remove the rule that lets cameras raise a province to WATCH
T6	Does low visibility at airports separate smoky from clean hours?	64% of hours under 10 km at 25–37.5 against 20% below 10; one airport supplies 76%	Gap of at least 20 points, cluster CI above zero, no airport over 30% of hours	Keep visibility as a reason on the card and never in the score
T7	Does success mean data?	Two token-gated sources logged 100% success and wrote no rows	Zero token-gated sources that skip silently	A skipped source is reported as "not configured" on the status page

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